

Signal Detection in the presence of AWGN

by Ameenudeen PE

GEC Idukki

June 11, 2020

- System Model
- Signal Space Concept-Review
- Signal Space Representation of AWGN.
- Continuous AWGN Channel to Vector channel conversion
- Detection in the presence of Noise
- Statistical Characterisation
- MAP & ML Rule
- Correlation Receiver
- Probability of Error

System Model

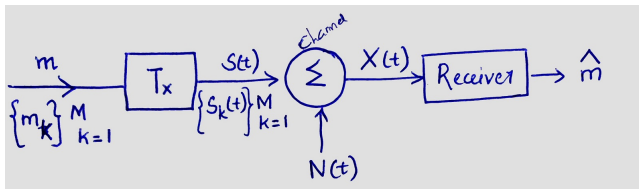


Figure: M-ary Communication Model

- We are concerned with the transmission of one of the 'M' messages.
- Message transmitted using waveforms $s_i(t)$, $i = 1, 2, \dots, M$
- $x(t)$ = Noisy version of the message signal
- **Receiver Goal:** To detect which one of the 'M' message signal was transmitted
- Channel is corrupted by noise
- Simplification: Use **Signal Space Concept** and convert signals to vectors

Signal Space Concept

- At the receiver $x(t) = s_k(t) + n(t)$
- Only $s_k(t)$ is deterministic. Noise is random and therefore $x(t)$.
- Model as RP: $X(t) = N(t) + s_k(t)$
- $N(t)$ is zero mean AWGN
- We know, $s_k(t) = \sum_{j=1}^N s_{kj} \phi_j(t)$, where $\{\phi_j(t)\}_{j=1}^N$ is the orthogonal basis set. where,

$$s_{kj} = \int s_k(t) \phi_j(t) dt \quad (1)$$

- Gram Schmidt process is invalid for Random Processes.
- The signal space expansion can be, $X(t) \stackrel{m.s}{=} \sum_{j=1}^{\infty} X_j \phi_j(t)$, where the convergence is in the mean square sense:
i.e.

$$E \left[\int |X(t) - \sum_{j=1}^{\infty} X_j \phi_j(t)|^2 dt \right] = 0 \quad (2)$$

- Under quite general condition a set of $\{\phi_j(t)\}$ can be found out and is called KL expansion. Each X_j is a random variable.

Signal Space Representation of AWGN

- Consider the WSS Noise process $N(t)$ modelled as zero mean AWGN.
- $E(N(t)) = 0$, $R_N(\tau) = \frac{N_0}{2}\delta(\tau)$
- Now Let $N(t) = \sum_{j=1}^{\infty} N_j \phi_j(t)$, where $N_j = \int_0^T N(t) \phi_j(t) dt$
- Now $E(N_j) = E\left(\int_0^T N(t) \phi_j(t) dt\right) = \int_0^T E(N(t)) \phi_j(t) dt = 0$

Consider covariance of N_k & N_j

- Considering Covariance

$$\begin{aligned}
 \text{Cov}(N_k, N_j) &= E(N_k, N_j) - E(N_k)E(N_j) = E(N_k, N_j) \\
 &= E \left[\int_0^T N(t)\phi_k(t)dt \int_0^T N(l)\phi_j(l)dl \right] \\
 &= \int_0^T \int_0^T E(N(t), N(l))\phi_k(t)\phi_j(l)dtdl \quad (3) \\
 &= \frac{N_0}{2} \int_0^T \int_0^T \delta(t-l)\phi_k(t)\phi_j(l)dtdl
 \end{aligned}$$

$$\text{Cov}(N_k, N_j) = \begin{cases} \frac{N_0}{2} & k = j \\ 0 & k \neq j \end{cases} \quad (4)$$

- Covariance=0 means random variables N_k, N_j are uncorrelated. Since the random process is Gaussian, N_k, N_j are also independent
- Consider the orthonormal basis function $\{\phi_j(t)\}_{j=1}^N$. Use this basis with approximate extension to represent the noise $N(t)$.

- Divide $N(t)$ into two components: $N(t) = \hat{N}(t) + N_1(t)$ where, $\hat{N}(t) = \sum_{j=1}^N N_j \phi_j(t)$

$$N_j = \int_0^T N(t) \phi_j(t) dt \quad (5)$$

$$N_1(t) = N(t) - \hat{N}(t) \quad (6)$$

- We know $X(t) = s_k(t) + N(t)$

$$X(t) = \sum_{j=1}^N s_{kj} \phi_j(t) + \sum_{j=1}^N N_j \phi_j(t) + N_1(t) \quad (7)$$

$$X(t) = \sum_{j=1}^N (s_{kj} + N_j) \phi_j(t) + N_1(t) \quad (8)$$

- Define

$$\begin{aligned} X_j = s_{kj} + N_j &= \int_0^T s_k(t) \phi_j(t) dt + \int_0^T N(t) \phi_j(t) dt \\ &= \langle s_k(t), \phi_j(t) \rangle + \langle N(t), \phi_j(t) \rangle \\ &= \langle (s_k(t) + N(t)), \phi_j(t) \rangle \end{aligned} \quad (9)$$

$$\implies X_j = \langle X(t), \phi_j(t) \rangle = \int_0^T X(t) \phi_j(t) dt \quad (10)$$

- Subst. back

$$X(t) = \sum_{j=1}^N X_j \phi_j(t) + N_1(t) = \hat{X}(t) + N_1(t) \quad (11)$$

where, $\hat{X}(t) = \sum_{j=1}^N X_j \phi_j(t)$.

- Received signal $X(t)$ is the sum of projection of $X(t)$ on $\{\phi_j(t)\}$ and $N_1(t)$
- $N(t)$ is independent of signal $s_k(t)$. Therefore $N_1(t)$ being a part of $N(t)$ will be independent of $s_k(t)$. Therefore no information about transmitted signal is there in $N_1(t)$.
- $N_1(t) = N(t) - \hat{N}(t)$ will be a Gaussian process. For any time t

$$\begin{aligned}
\text{Cov}[N_j, N_1(t)] &= E[N_j, N_1(t)] - E[N_j]E[N_1(t)] \\
&= E[N_j, N_1(t)] = E[N_j, N(t)] - E[N_j, \hat{N}(t)] \\
&= E[N(t), \int_0^T N(s)\phi_j(s)ds] - E[N_j, \sum_{l=1}^N N_l\phi_l(t)] \\
&= \int_0^T E[N(t), N(s)]\phi_j(s)ds - \sum_{l=1}^N E[N_j, N_l]\phi_l(t) \\
&= \frac{N_o}{2} \int_0^T \delta(t-s)\phi_j(s)ds - \frac{N_o}{2}\phi_j(t) = \frac{N_o}{2}\phi_j(t) - \frac{N_o}{2}\phi_j(t) = 0
\end{aligned} \tag{12}$$

- Thus $N_1(t)$ is uncorrelated and independent with all N_j 's and therefore it is independent of $\hat{N}(t)$. Thus $N_1(t)$ is irrelevant and can be neglected for signal detection.

Continuos AWGN Channel to Vector channel conversion

Conventions used

$$X(t) \rightarrow \hat{X}(t) : \underline{X} = [X_1, X_2, \dots, X_N], X_j = \int_0^T X(t) \phi_j(t) dt$$

$$s_k(t) : \underline{S}_k = [s_{k1}, s_{k2}, \dots, s_{kN}], s_{kj} = \int_0^T s_k(t) \phi_j(t) dt$$

$$N(t) : \underline{N} = [N_1, N_2, \dots, N_N], N_j = \int_0^T N(t) \phi_j(t) dt$$

- $\hat{X}(t)$ contains the desired signal wave and the projection of the channel noise on the N-dimensional signal space spanned by $\{\phi_j(t)\}$.
- For the design of optimal detector the continuos AWGN channel of the form $X(t) = s_k(t) + N(t)$ is converted to an equivalent N-dimensional vector channel $\underline{X} = \underline{S}_k + \underline{N}$.

- $\underline{N} : (N_j)_{j=1}^n$ is N IID Gaussian Random variables with mean $E[N_j] = 0$ and variance $E[N_j^2] = \frac{N_0}{2}$
- PDF of $\underline{N}$

$$P_{\underline{N}}(n) = \prod_{j=1}^N \frac{1}{\sqrt{2\pi \frac{N_0}{2}}} e^{-\frac{(n_j)^2}{2 \frac{N_0}{2}}}$$

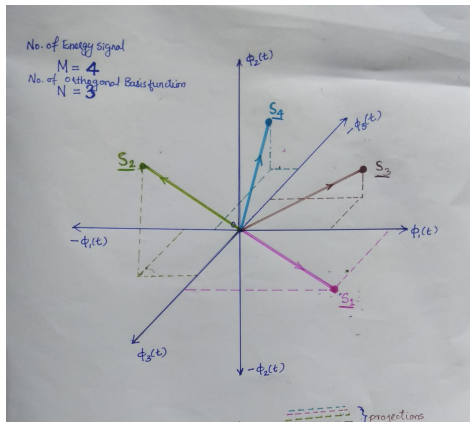
$$P_{\underline{N}}(n) = \frac{1}{(\pi N_0)^{\frac{N}{2}}} e^{-\frac{\|n\|^2}{N_0}} \quad (13)$$

where

$$\|n\| = \sqrt{n_1^2 + n_2^2 + \dots + n_N^2}$$

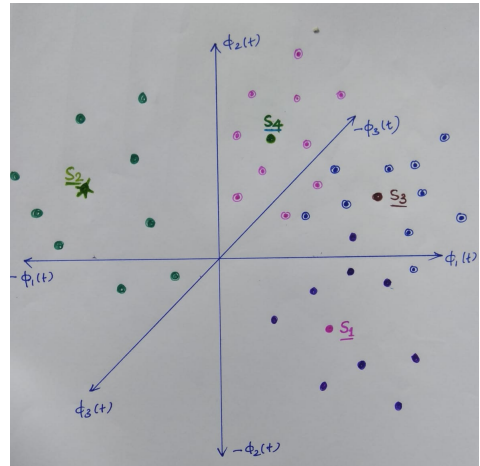
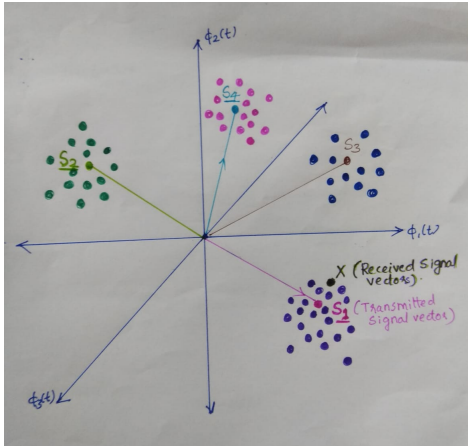
is the sample length of noise vector in the hyperspace.

- $\underline{X} = \underline{S}_k + \underline{N}$. Since noise distribution has a spherical symmetry distribution of $\underline{X}$ is also spherical around the point $\underline{S}_k$. A scatter region is formed around the signal point $\underline{S}_k$
- There will be M scatter regions around $\underline{S}_1, \underline{S}_2, \dots, \underline{S}_M$ corresponding to each of the M input messages.
- By analysing the position of vector $\underline{X}$ we can decide on the transmitted message with a small probability of error P_e .



Constellation Diagram

- In principle each scatter region extends upto infinity. But the probability of observing the received signal diminishes rapidly as the point is scattered away from the centre



Detection in AWGN

If we find a vector $\underline{X}$ closer to $\underline{S}_j$ there is a probability that it has been generated by some other message vector $\underline{S}_k$. This may result in an error. The task of optimum receiver is to decide from $\underline{X}$ which message was transmitted with minimum P_e

Divide the entire signal space into M disjoint regions $Z_1, Z_2 \dots Z_M$. If $\underline{X}$ falls into the decision region Z_k we will decide that message transmitted is m_k . Thus the problem of designing the receiver reduces to choosing the boundaries of decision regions $Z_1, Z_2 \dots Z_M$.

Statistical Characterisation

- The conditional probability density functions $f_{\underline{X}}(\underline{x}/m_j), j = 1, 2, \dots, M$ are the most important characteristic of an AWGN channel. This leads to a functional dependence on the $\underline{X}$ given that m_j was transmitted using $s_j(t)$
- $f_{\underline{X}}(\underline{x}/m_j)$ is also called as the likelihood function.
- We know $\underline{X} = \underline{S}_j + \underline{N}$. $\underline{S}_j$ is a constant and the noise vector is random. So obviously $\underline{X}$ will have the same distribution as $\underline{N}$ but centered (mean) at the point $\underline{S}_j$.
- By using the expression for PDF of $\underline{N}$,

$$f_{\underline{X}}(\underline{x}/m_j) = \frac{1}{(\pi N_0)^{\frac{M}{2}}} e^{\frac{-\|\underline{x} - \underline{s}_j\|^2}{N_0}} \quad (14)$$

- In practice we use the log likelihood function defined by $\log(f_{\underline{X}}(\underline{x}/m_j))$ since log is a monotonically increasing function.

Coherent Detection

The Problem

Given an observation vector $\underline{X}$ perform a mapping from $\underline{X}$ to an estimate $\hat{m}$ of the transmitted symbol m_j in a way that would minimize the probability of error, P_e .

$$\text{Here } P_e = P(m_j \text{ not sent} / \underline{X}) = 1 - P(m_j / \underline{X})$$

Optimal Decision Rule(MAP)

Set $\hat{m} = m_j$ if,

$$P(m_j / \underline{X}) \geq P(m_k / \underline{X}) \quad \forall k \neq j, k = 1, 2 \dots M$$

- The above rule can be reframed using Bayes theorem as:

$$\hat{m} = \underset{1 \leq j \leq M}{\operatorname{argmax}} P(m_j / \underline{x}) = \frac{P(m_j) f_{\underline{X}}(\underline{x} / m_j)}{f_{\underline{X}}(\underline{x})} \quad (15)$$

- The denominator term $f(\underline{x})$ is common for all j and therefore can be neglected.

$$\hat{m} = \underset{1 \leq j \leq M}{\operatorname{argmax}} \left[P(m_j) \frac{1}{(\pi N_0)^{\frac{N}{2}}} e^{-\frac{\|\underline{x} - \underline{s}_j\|^2}{N_0}} \right] \quad (16)$$

- Taking log and removing irrelevant constant terms,

$$\begin{aligned} \hat{m} &= \underset{1 \leq j \leq M}{\operatorname{argmax}} \left[\ln P(m_j) - \frac{\|\underline{x} - \underline{s}_j\|^2}{N_0} \right] \\ &= \underset{1 \leq j \leq M}{\operatorname{argmax}} \left[\frac{N_0}{2} \ln P(m_j) - \frac{1}{2} [\|\underline{x}\|^2 + \|\underline{s}_j\|^2 - 2\underline{x} \cdot \underline{s}_j] \right] \end{aligned} \quad (17)$$

- $\hat{m} = \underset{1 \leq j \leq M}{\operatorname{argmax}} \left[\frac{N_0}{2} \ln P(m_j) - \frac{1}{2} E_j + \underline{x} \cdot \underline{s}_j \right]$, where E_j is the energy of $s_j(t)$ and $\|\underline{x}\|^2$ is neglected.

MAP Detector for AWGN

$$\hat{m} = \underset{1 \leq j \leq M}{\operatorname{argmax}} [c_j + \underline{x} \cdot \underline{s}_j] \text{ where } (c_j = \frac{N_0}{2} \ln P(m_j) - \frac{1}{2} E_j) \text{ is constant for a particular 'j'}$$

- Consider all the messages are equiprobable , $P(m_j) = \frac{1}{M}$

Maximum Likelihood (ML) Detection

$$\begin{aligned}\hat{m} &= \underset{1 \leq j \leq M}{\operatorname{argmax}} \left[\frac{N_0}{2} \ln P(m_j) - \frac{1}{2} \|\underline{x} - \underline{s}_j\|^2 \right] \\ &= \underset{1 \leq j \leq M}{\operatorname{argmax}} \left[-\frac{1}{2} \|\underline{x} - \underline{s}_j\|^2 \right]\end{aligned}\quad (18)$$

ML Detector for AWGN

$$\hat{m} = \underset{1 \leq j \leq M}{\operatorname{argmax}} [f_{\underline{x}}(\underline{x}/m_j)] = \underset{1 \leq j \leq M}{\operatorname{argmin}} [\|\underline{x} - \underline{s}_j\|^2]$$

ML detector reduces to minimum Euclidean distance decoder.

Geometric Interpretation

Receiver receives the vector $\underline{x}$ and looks for all $\underline{s}_j$'s that is closest to it using standard Euclidean distance. Such a detector is called minimum distance /nearest neighbour detector.

ML Decoder

- Divide the signal space into M disjoint regions Z_k , $k = 1, 2, \dots, M$
- The observation vector $\underline{x}$ lies in the region Z_j if the Euclidean distance $\|\underline{x} - \underline{s}_k\|$ is minimum for $k = j$.
- The observation vector $\underline{x}$ lies in the region Z_j if $\sum_{i=1}^N (x_i - s_{ki})^2$ is minimum for $k = j$.

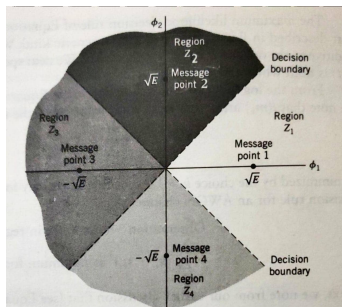


Figure: Decision Regions

MAP Detector- Implementation

$$\hat{m} = \underset{1 \leq j \leq M}{\operatorname{argmax}} [c_j + \underline{x} \cdot \underline{s}_j], \quad x_j = \int_0^T x(t) \phi_j(t) dt \quad (19)$$

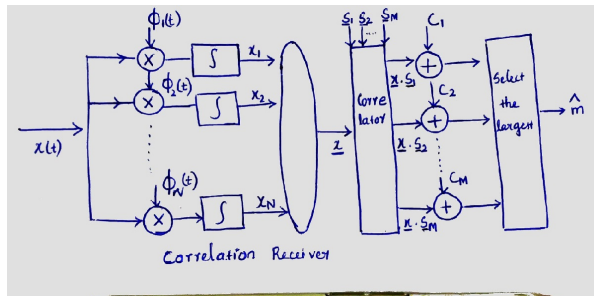


Figure: Correlation Receiver

- For ML detector $c_j = -\frac{1}{2}E_j$

Probability of Error in Correlation Receiver (ML)

- Suppose that a symbol m_j is transmitted and $\underline{x}$ is received. An error occurs if $\underline{x}$ does not fall in Z_j
- Averaging over all symbols the average probability of error

$$P_e = \sum_{j=1}^M (P(m_j) P(\underline{x} \text{ does not lie in } Z_j | m_j)) \quad (20)$$

$$= \frac{1}{M} \sum_{j=1}^M (P(\underline{x} \text{ does not lie in } Z_j | m_j)) = 1 - \frac{1}{M} \sum_{j=1}^M (P(\underline{x} \text{ lies in } Z_j | m_j))$$

Probability of Error

We get it as

$$P_e = 1 - \frac{1}{M} \sum_{j=1}^M \int_{Z_j} f_{\underline{X}}(\underline{x} | m_j) d\underline{x}$$